

Ali Abid

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EDUCATION

Queen's University

Bachelor of Applied Science, Major in Computer Engineering (3.9 GPA)

Kingston, ON

Sep. 2025 – Apr. 2030 (Expected)

TECHNICAL SKILLS

Languages: C/C++, Assembly (x86, arm64), Rust, Java, C#

Frameworks: Arduino, STM32, Raspberry PI

Tools: VSCode, Arduino IDE, Altium, STM32CubeMX, Visual Studio, PowerBI

EXPERIENCE

Low Voltage Team Member

Queen's Formula SAE

Sep. 2025 – Present

Kingston, ON

- Engineered a custom vehicle dashboard by integrating an **STM32** microcontroller with an LCD display to provide real-time telemetry and performance feedback for the racing team.
- Designed and routed a multi-layer custom PCB using **Altium Designer**, ensuring optimal signal integrity and a compact form factor for high-vibration automotive environments.
- Validated hardware reliability by developing an **Arduino**-based testing suite to prototype display logic and sensor integration before finalizing the production-grade PCB.

Assitant Teacher

ICM — SteamOji

Aug. 2024

Markham, ON

- Facilitated technical instruction on embedded logic and control flow by guiding students through **Python** and Scratch to develop autonomous behaviors for **robotic** platforms.
- Mentored youth in hardware-software integration using **Microsoft microbit** controllers, teaching fundamental concepts in sensor data acquisition and peripheral interfacing.
- Guided rapid prototyping and mechanical design using TinkerCAD and **3D printing** to manufacture custom structural components for student-led engineering projects.

PROJECTS

Water Filtration Prototype | Arduino, C++, ADAFruit

Sep. 2025 – Dec. 2025

- Programmed an embedded control system on an **Arduino Mega** to automate pumps, motors, and turbidity sensors for a **multi-stage** water filtration process in a team environment.
- Implemented non-blocking timing logic using `millis()` to coordinate system operations, ensuring **accurate** process timing while allowing an emergency stop to interrupt execution at any time.
- Debugged** low-level **Arduino** issues, including resolving integer overflow errors affecting timing calculations, improving system **correctness** and **reliability**.
- Designed** and printed a mechanical dowel stabilizer and wired all electrical components using a motor shield, **integrating** hardware and software into a functional prototype.

Line Following Robot | Arduino, C++

Feb. 2024

- Developed an **autonomous** line-following robot using **Arduino Uno** and infrared sensors for high-speed navigation.
- Integrated ultrasonic sensors for real-time obstacle detection and **automated** safety braking logic.
- Calibrated IR sensor thresholds through iterative testing to improve line-tracking **reliability** on complex curves.
- Optimized motor control parameters to maintain path **stability** and prevent derailment during high-speed operation.

LEADERSHIP

VESPA Robotics

Sep. 2024 – Feb. 2025

- Mentored new members on robot design and system integration to streamline team onboarding and technical proficiency.
- Coordinated team operations and strategic planning to drive peak performance during competitive events.